Mikhail Razakov

Algorithm / Solver Engineer

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I build reusable algorithmic components: AdvancedAlgorithms, PS-form memory-dependence analysis, and graph infrastructure for an LLVM-track internship.

12 C++23 modules
graphs, trees, strings, and data structures

1st · 104/104
PS-form; only maximum result among 49

500k vertices
stress tests for recursion and accidental $O(n^2)$

EXPERIENCE

2026 — PRESENT

AdvancedAlgorithms

- Explicit contracts, complexity notes, differential oracles, and structural invariants.
- ASan/UBSan and separate correctness/performance suites.

JUL–AUG 2026

MCST

- Shared graph model plus RPO/cycles, Dijkstra, Tarjan SCC, and Dinic.

SELECTED PROJECTS

PS-form Analyzer

Conservative memory-dependence analysis with exact and metamorphic testing.

Wist SSA

Callable-first SSA, CFG/dominance verification, and descriptor-driven folding.

x86 Codegen Lab

Liveness, live intervals, and linear-scan register allocation.

SKILLS

Algorithms
graphs, trees, strings, data structures, optimization

Correctness
differential tests, exact oracles, invariants, sanitizers

Complexity
explicit bounds, large inputs, non-flaky performance gates

Languages
C++23, C17, Python, C#/.NET, Rust

EDUCATION

Software Engineering student at HSE University

RECOGNITION

Ranked 1st of 49 in MCST selection; HSE Vysshaya Proba prize-winner; top MEPhI Junior results in 2025/2026.

TECHNICAL COMMUNICATION

Teaching algorithms and reviewing solutions since 2021.

AVAILABILITY

Moscow / remote · up to 20 hours/week from September 2026